

# Ibn Turka's (d. 835/1432) Pythagorean Elevation of Sight over Hearing as Vehicle of Total Perception

Matthew Melvin-Koushki

## 1. Introduction

Occult philosopher *par excellence* of the Timurid Empire (r. 771-912/1370-1507), Ibn Turka gorgeously synthesized a new-old Pythagorean physics-metaphysics that would do much to shape everything from imperial ideology and historiography to science and technology to literary, visual and architectural culture throughout the vast early modern Persian cosmopolis. Lettrism (*'ilm al-ḥurūf*) was Doctor Littleturk's preferred vehicle to that universalist, cosmic-encyclopedist end.<sup>1</sup> Peculiar to the Hellenic-Judeo-Islamic Mediterranean zone, this Biblical-Qur'ānic alphanumeric science was Arabic twin sister to Hebrew kabbalah, whose eventual and garbled western European reception is a major structural element of our orthodox Eurocentric "Scientific Revolution" narrative, whereby the world came to be both visually cataloged and mathematically calculated as an algorithmic Book by Latin Christians only—the genesis of the modern Panopticonic constitution.<sup>2</sup> To the contrary, Ibn Turka—a Muslim Iamblichus and Cusanus and Bruno and Pico and Dee, but considerably more influential than these peers until the colonial period—was the first thinker in the Greater Western (Hellenic-Judeo-Christo-Islamic) tradition to develop a specifically Pythagorean-Platonic mathematical-theurgic philosophy that

---

<sup>1</sup> "Doctor Littleturk" captures his status as the most prominent and influential member of the Turka (lit., Little-turk) line, originally from Khujand in what is now Tajikistan, and much like the more infamous Welshman Doctor Dee (d. 1608) a court sage and mage responsible for the first occult-scientific theorization of Timurid imperialism (Melvin-Koushki, "Love").

<sup>2</sup> The Greek alphabet appears to have been the first to be systemically used alphanumerically, whereby letters are numbers too, and following it Hebrew and especially Arabic, whose 28 letters, uniquely, encode the units 1-9 and the orders of magnitude 10, 100 and 1,000 (Acevedo). This cosmic marriage of letter and number made both kabbalah and lettrism the organic vehicle of various early modern forms of Pythagoreanism, a primary driver in turn of the projects of such Saints of Science as Francis Bacon (d. 1629), Johannes Kepler (d. 1630) and Isaac Newton (d. 1727) to definitively mathematically *read and rewrite the world*. Due to persistent historiographical colonialism and Orientalism, however, major, self-styled Pythagorean philosophers like Ibn Turka have been erased from this story for the sin of being non-European, despite the fact that they too pushed ardently and successfully for the mathematization of the cosmos (Melvin-Koushki, "Scientific Revolution," "Grammatology"). On the Two Books doctrine in the history and historiography of science see Killeen and Forshaw.

could scientifically, politically and culturally challenge the hegemonic Avicennan Aristotelian-Platonic hybrid system.<sup>3</sup> Among other feats, it turned Ibn Sīnā's (Avicenna, d. 427/1037) sensory theory quite literally, or rather *lettrarily*, on its head.

That theory, semi-Platonic, classifies the senses along a strictly linear axis that ascends from gross materiality to purest metaphysical abstraction. Grosset of all physical senses is touch, and noblest is sight, the most abstract and hence bridge to the rational and imaginal faculties, far more abstract still and so more preferred. Suhrawardī's (d. 563/1191) Illuminationist or Enlightenment (*Ishrāqī*) update to the Avicennan system, while valorizing Sight as portal of Light and more explicitly planetary-theurgic, still demanded sensorial Ascent (*mi'rāj*) from shadowy touch to solar Mind as an epistemic imperative.<sup>4</sup> But Ibn Turka's Pythagorean-Prophetic retort demands sensorial Descent (*tanzīl*) too to create a total existential Circulation through the Marriage of All Opposites in classic Hermetic *as above so below* mode—with equal emphasis on the *below*.<sup>5</sup> Like his junior Latin Christian contemporary and fellow judge Nicholas of Cusa (d. 1464), now famed for coining the term *coincidentia oppositorum* (Albertson), our Timurid worthy posited the *majma' al-aḍḍād* as the basis of a new mathematical philosophy.<sup>6</sup> Unlike Cusanus, who as a fellow Pythagorean also rejected the Aristotelian hierarchy of the senses by declining to privilege any, he argued for sight as the most Descended of all physical senses: it alone is able to perceive all contradictory data At Once (*tawhīd*), and is hence both cosmic terminus of and elevator back to mathematical Mind, equally solar and lunar and through Number also encompassing of all opposites. For the stuff of material being is immaterial Light—which Mind-bendingly most totally gravitationally manifests through the Black Hole that is the calligraphed Letter. Algorithmically but also visually *reading* darkest and grossest physical reality itself, Plato's Cave, the Second Book, turns out to be the best Wormhole to the One.<sup>7</sup>

<sup>3</sup> The celebrated Syrian Pythagorean-Platonist philosopher Iamblichus (d. 325 CE)—whose disciples were responsible for designing the Hagia Sophia, an architectural and occult-technological feat with which the Tāj Maḥal quite consciously rhymes—here provides the most salient late-antique precedent for Ibn Turka's own post-Avicennan imperial system, pivoting on *godwork* (theurgy as theosis) and planetary ascent by means of ritual mathematical language. See Adams, Shaw.

<sup>4</sup> Ibn Turka's use of *qalb* (heart) here roughly corresponds to Greek *nous*, a slippery concept variously translated or approximated by various arabophone philosophers, and here rendered as "Mind." By the same token, I translate *Ishrāq* as "Enlightenment" to better convey its radical yet systematically philosophical focus on Light as the nonmaterial ground of being—thereby helping inspire Ibn Turka's valorization of Sight as vehicle to Mind. On Suhrawardī's planetary theurgy see Piątak.

<sup>5</sup> In other words, it is a classic panpsychist critique of Platonic idealism, which tends to champion Ascent alone as escape from the Cave. On Hermetic Ascent-and-Descend see Hanegraaff.

<sup>6</sup> He also uses terms like *ta'ānuq al-aṭrāf*, *ta'ānuq ḍiddayn*, *majma' li-l-ṭarafayn* and *mu'tanaq li-l-mutaqābilayn* to signify, somewhat like Yin-Yang, the perpetual cosmic marriage of opposites.

<sup>7</sup> While "Black Hole" and "Wormhole" are obviously not terms Ibn Turka uses, they conceptually accurately convey something of his physics-metaphysics, whereby the pitch-blackness of ink represents the bottom of the gravity well that is Love, which eternally coalesces all worlds—and yet this sheerest Blackness is through Sight and therefore Mind the magic door back to celestial Light (Melvin-Koushki, "Grammatology"). Speaking of astrophysics, that the founder of the Turka line emigrated to Isfahan from Khwarazmshahid Khujand would

The effects of Ibn Turka's radically progressivist-cum-traditionist occult upsetting of the Avicennan-Aristotelian sensory applet manifest on the Worldpage already within his lifetime, especially on the scientific, political and book-arts fronts (all the same to an imperial lettrist). Scientifically, it is no accident that the publication of his magnum opus, *Investigations* (*Kitāb al-Mafāḥiṣ*), the first summa of Islamic Pythagoreanism, coincided with Ulugh Beg's (r. 811-53/1409-49) founding of the Samarkand observatory in 823/1420.<sup>8</sup> There he follows the original Brethren of Purity in declaring Number the true source of all Science, but goes much further in constellating a total lettrary, empiricist cosmology—one which demands direct *visual* observation as the only route to the *mathematical* precision necessary to read and write the cosmos.<sup>9</sup> So inspired, Ibn Turka's astronomer friends launched an epochally successful observational program, resulting in the *New Sultanic Star Tables* (*Zīj-i jadīd-i sulṭānī*), the most mathematically accurate catalog of the heavens ever produced to date. The first in the world to achieve arcminute precision, this Timurid *zīj* not only completely supplanted both Ptolemy's *Almagest* and the theoretically more advanced *Ilkhanid Star Tables* (*Zīj-i Īlkhānī*), but remained the model authority throughout Afro-Eurasia well after the emergence of telescopic observation programs from the eleventh/seventeenth century onward; even in the thirteenth/nineteenth century it was still being used everywhere from Delhi to London.

By the same token, Ibn Turka's first Timurid patron, Iskandar Sulṭān (r. 812-17/1409-14), grandson of Temür and ambitious ruler of Fars, was so inspired as to make Pythagorean astro-lettrism the basis of his bid for Alexandrian empire.<sup>10</sup> That bid failed; yet his magnificent illustrated horoscope remains to this day the most iconic surviving Islamic specimen of the genre ever produced, an ultimate visual epitome of Platonic saint-philosopher-kingship. Now ubiquitously reproduced on book covers, conference posters and websites to represent "Science in Islam," regardless of period or place, this specifically Timurid Iranian testament to the intimate relationship of the boom in Persianate book arts to Ibn Turka's Pythagorean Book Science can only be accurately Hermetically riddled by lettrist means.<sup>11</sup> That intimate relationship is further

---

seem to be another of those weird historical rhymes, given the derivation of "algorithm" from the origin name of Muḥammad b. Mūsā al-Khwārazmī (d. ca. 233/850), the famed 'Abbasid mathematician, astronomer and geographer, who also penned an important West-East *zīj*.

<sup>8</sup> Ibn Turka gifted and dedicated another major lettrist work, *Sharḥ al-basmala* (*Commentary on the Basmala*), to the same ruler six years later, there explicitly styling him a Pythagorean-Platonic philosopher-king (Melvin-Koushki, "Love," "Sultan-Scientists").

<sup>9</sup> See Ikhwān al-Ṣafā', *Rasā'il*, 1:21-22; Melvin-Koushki, "New Brethren." A leading figure in the New Brethren of Purity network of the ninth/fifteenth century and after, explicitly modeled on its fourth/tenth-century Pythagoreanizing namesake, Ibn Turka nevertheless breaks sharply with their standard Aristotelian hierarchy of the senses (on which see *ISH*, vol. 2, ch. 18).

<sup>10</sup> Alexander the Great, long the classic icon of West-East empire, became a favorite early modern symbol of Turko-Mongol Perso-Islamic occult-scientific imperial universalism as developed by the Timurids in particular (Melvin-Koushki, "Sultan-Scientists").

<sup>11</sup> London, Wellcome Collection, MS Persian 474 (painted 1411 CE); its current London housing has not a little to do with its status as colonial icon. Despite its celebrated iconicity, art historians have yet to riddle this kaleidoscopically and psychedelically visual-lettrist talisman of empire, whose twinings within twinings within more Hermetic twinings still are designed as a map for the Platonic-Alexandrian Persian-Mongol conquest of

confirmed by Ibn Turka's own with the Timurid prince Bāysunghur (d. 837/1433), like his cousin Iskandar famed as a patron of the book arts, but also an accomplished calligrapher in his own right; his status as visionary patron of the exceptionally monumental Bāysunghur Qurʾān—another quintessential imperial talisman whose mere fragments are much coveted by collectors today—is likewise no accident.<sup>12</sup>

A Qurʾān-cosmographer by definition, but no otherworldly mystic, Ibn Turka immortalizes these contemporary visual-mathematical but also very manual achievements with the Qurʾānic phrase *men of main and vision* (Q 38:45). The mainstreaming of his Pythagorean world-writing, ontocalligraphic imperialism during the tenth/sixteenth century thus informed unprecedented and explicitly psychedelic (Mind-manifesting) forms of painting and architecture and Riddle culture generally throughout the Persianate world. Two world-famous triumphs in particular incarnated his quite literally *visionary* program in stone: the Safavid transformation of Isfahan into the world's first fully Pythagorean-panacean capital, and the Mughal (Indo-Timurid) construction of the Tāj Maḥal as mathematical model of Heaven on Earth and Platonic spaceship to human transplanetary perfection. That these ontogrammatological architectural technologies have been colonially reduced to hopping tourist hotspots only testifies to their ensorcelling power as portals to the cosmic sensorium still—even as the almighty Algorithm seeks to rewrite the world again (Melvin-Koushki, “Grammatology,” *On Doffing*; Koch; Emami).

In strictly literary, non-ontogrammatological terms, however, Ibn Turka's valorization of sight over hearing was hardly unprecedented; al-Jāḥiẓ himself had already done the same as living epitome of the book revolution almost six centuries before (Montgomery).<sup>13</sup> But many Islamicate cultures, possessed of *munāẓara* consciousness, were so productively addicted to ambiguity that the venerable Eye *versus* Ear Debate still had no clear victor by the turn of the ninth/fifteenth century; each contestant won with roughly the same frequency depending on the traditionist or literary predilections of a given author (Lange). And some, like Abū l-Majd Tabrīzī's (d. after 736/1336) entry, presided over by the Heart, simply end in a draw.<sup>14</sup> Ibn Turka is the first Muslim man of letters to take up that Qurʾānic sensory trinity Ear-Eye-Heart (Q 16:78, etc.)—also deployed by the traditionist Ibn Qayyim al-Jawziyya (d. 751/1350),<sup>15</sup> sworn enemy to the Pythagoreans (*fīthāghūriyyūn*) as the spiritual cause of the Mongol conquest—to have Heart declare Eye the winner within not only a systematic Pythagorean framework but a gorgeous literary one too, magnifying its intellectual, sociopolitical and indeed aesthetic and material-cultural impact everywhere from the Bosphorus Strait to the Yamuna River and beyond.

To be a lettrist is usually to be a litterateur and radical humanist too (Melvin-Koushki, “Sex”). No mere philosopher, Ibn Turka has likewise been acclaimed the greatest Persian prose

---

the cosmos and history itself: the clue is in the crowns. I riddle it in Melvin-Koushki, “Selenocentrism.”

<sup>12</sup> These continue to be auctioned off at eye-watering prices by houses like Sotheby's as we speak, where recent successful bids for even partial single lines range from a mere £103,250 to well over £500,000!

<sup>13</sup> On al-Jāḥiẓ's sensorium, see *ISH*, vol. 2, ch. 7.

<sup>14</sup> On Tabrīzī's debate, see *ISH*, vol. 2, ch. 13.

<sup>15</sup> On Ibn al-Qayyim's sensorium, see *ISH*, vol. 2, chs. 32 and 45.

stylist of the fifteenth century; true to Pythagorean form, that style is masterfully prosimetric.<sup>16</sup> He launched his writing career during his fifteen-year tenure in Mamluk Cairo (ca. 795-810/1393-1408), where he trained in law, Arabic letters and occult science—and made mortal enemies at court, most infamously Ibn Khaldūn (d. 808/1406), a significant fraction of whose *Prolegomenon* (*Muqaddima*) is a violently puritanically anti-philosophy, witchhunting reaction to the contemporary occult-scientific revolution (Melvin-Koushki, “Scientific Revolution”). The very year of his enemy’s death, Ibn Turka published an ornate Persian prosimetric cycle of *Five Debates* (*Munāẓarāt-i khams*): that between the faculties of Sight (*baṣar*) and Hearing (*samʿ*) crucially supports the triumph of Love (*ʿishq*) over mere Reason (*ʿaql*) as King (*sulṭān*) of the cosmic sensorium.<sup>17</sup> Formally a retort (*javāb*) to literary debates like Abū l-Majd Tabrīzī’s, here Sight gives Hearing a sound drubbing—a drubbing made all the more scientifically conclusive with lettrist proofs advanced in many subsequent works. The closing and winning graphophile argument pivots, ironically, on this fatal traditionist thrust against aurophile traditionism itself: “Make specious arguments all you like—but haven’t you heard the saying ‘Believe what you see, not what you only hear’? Or the hadith ‘True is the eye—but hearing’s a liar’?”<sup>18</sup>

But it is also formally a philosophical retort to both Ibn Sīnā and Suhrawardī. In an equally deliciously ironic mic-drop, Sight clinches its claim and ends the debate with this Suhrawardī quatrain:

I plead with my neighboress, tears flowing:  
I’m resolved to leave this province!  
Let me go, with no regrets—  
For the noblest of stars are those that travel.<sup>19</sup>

This Enlightenment injunction to Soul (*nafs* as neighboress) is classically Platonic-theurgic. But here it is Sight that is her vehicle of Ascent to the One via the visible planetary spheres—and Decent back from the One too: it alone among the physical senses can *travel* and *telescope* all realms. This is the triumph of Eye over Ear as universal aperture of Mind, the only two faculties to Marry All Opposites by cowering them across pagescapes and landscapes and starscapes simultaneously.

<sup>16</sup> Specifically, his Arabic works are theoretically and hence stylistically dense, while his more accessible Persian works combine exquisite rhymed prose with nonrhymed but still wonderfully mellifluous prose studded with lines of Persian and Arabic poetry, perfectly metric. See Bahār, vol. 3, pp. 223-34, 352.

<sup>17</sup> Cf. Isaac Newton as an equally committed Pythagorean occultist on Gravity as Love. On this concept’s scientific, technological, political and ethical stakes see Melvin-Koushki, “Love.”

<sup>18</sup> Ibn Turka, *Munāẓarāt-i khams*, p. 85: *al-ʿayn ṣādiqa wa-l-samʿ kadhdhāb*. Ibn Turka first cites this hadith in *ShND* (p. 60), following the classic Ibn ʿArabī-school commentators on Ibn al-Fāriḍ’s *Ode in T*, ʿIzz al-Dīn Kāshānī (d. 735/1335) and Saʿīd al-Dīn Farghānī (d. 700/1301)—but here uses it to a strikingly new scientific-imperial purpose: the irony in recording oral statements in writing to disparage orality is naturally quite deliberate.

<sup>19</sup> *Ibid.*, p. 86.

I translate below sections from three of Ibn Turka's most seminal works that best illustrate his watershed and wonderfully lettrary Pythagorean valorization of sight over hearing, proceeding in chronological order. First is his *Commentary on Stringing the Pearl* (*Sharḥ-i Naẓm al-durr*), a gorgeously prosimetric Persian commentary on the famous *Ode in T* (*al-Tā'īyya al-kubrā*) of Ibn al-Fāriḍ (d. 632/1235), a primary teaching text in the Ibn al-ʿArabī school tradition; completed in Cairo in 806/1404, it channels Mamluk models to definitively poeticize and canonize Akbarian jazz philosophy for a Timurid courtly audience.<sup>20</sup> Also prosimetric but more programmatically scientific, his Persian treatise *On the Letters* (*Risāla-yi ḥurūf*, 817/1414) did much to define the explicitly astro-lettrist imperial program of Iskandar Sulṭān, while also inspiring both Bāysunghur and Ulugh Beg and various court scientists, book artists and humanists thereafter. That program is most definitively stated in his Arabic *Investigations* (*Maḥāḥiṣ*, 823-28/1420-25), one of the most materially influential philosophical manifestos in the whole of the Western tradition; impetus for the radically empiricist observational-mathematical program at Samarkand and anthem for the boom in Persianate book arts and architecture, it would continue to be avidly read by leading Safavid, Mughal and Ottoman and Shi'i and Sunni scholars alike, who made the Two Books doctrine (Qur'ān and Cosmos) scientifically, technologically, politically, culturally and indeed sensorially hegemonic across the Afro-Eurasian ecumene. In the translated section, Ibn Turka, a committed Sunni imamophile, champions Ibn al-ʿArabī's (d. 638/1240) infamous argument for the superiority of sainthood (*walāya*) to prophethood (*nubuwwa*) to fuse Imam ʿAlī with Pythagoras of Samos as the co-inventors of lettrism—and hence of literacy itself, both cosmic and pedestrian, manifestly predicated on the occult supremacy of sight.<sup>21</sup>

## References

- Acevedo, Juan, *Alphanumeric Cosmology from Greek into Arabic: The Idea of Stoicheia through the Medieval Mediterranean*, Tübingen: Mohr Siebeck, 2020.
- Adams, Peter Mark, *Hagia Sophia / Sanctum of Kronos: Spiritual Dissent in an Age of Tyranny*, London: Scarlet Imprint, 2023.
- Albertson, David, *Mathematical Theologies: Nicholas of Cusa and the Legacy of Thierry of Chartres*, Oxford: Oxford University Press, 2014.
- Bahār, Muḥammad-Taḳī (Malik al-Shu'arā'), *Sabk-shināsī, yā Tārīkh-i taṭavvur-i nathr-i fārsī*, 3 vols., Tehran: Zuvvār, 1381 Sh./2002.

<sup>20</sup> Among other unprecedented virtues, it is the first to quote Ḥāfiẓ (d. ca. 792/1390), Tongue of the Occult (*lisān al-ghayb*) himself—in a physics context no less. And speaking of jazz: that the radically free-playing, Mind-manifesting stylings of Sa'd al-Dīn Ḥamūya (d. 649/1252)—with Ibn al-ʿArabī (d. 638/1240) Ibn Turka's favorite lettrist source—riff on the visual and the aural simultaneously but in an explicitly *musical* mode makes the Mongol-Islamicizing Sufi no less Pythagorean. On Ḥamūya and jazz philosophy see Uy.

<sup>21</sup> The Letter as visual manifestation of occult Number whose organ is Mind (Heart) is encapsulated in the New Brethren of Purity's central motto: *He is the First and the Last, the Manifest and the Occult* (Q 57:3)—likewise the basis of Ibn Turka's radical new political philosophy (Melvin-Koushki, "Love," 66, 84).

- Emami, Farshid, *Isfahan: Architecture and Urban Experience in Early Modern Iran*, University Park: The Pennsylvania State University Press, 2024.
- Hanegraaff, Wouter J., *Hermetic Spirituality and the Historical Imagination: Altered States of Knowledge in Late Antiquity*, Cambridge: Cambridge University Press, 2022.
- Homerin, Th. Emil, *ʿUmar Ibn al-Fāriḍ: Sufi Verse, Sainly Life*, New York: Paulist Press, 2001.
- Ibn al-Fāriḍ, *al-Tāʾiyya al-kubrā*, in *Dīwān*, ed. Giuseppe Scattolin, Cairo: IFAO, 2004.
- Ibn Turka, *ʿAql va ʿishq, yā Munāẓarāt-i khams*, ed. Akram Jūdī-Niʿmatī, Tehran: Mīrās-i Maktūb, 1375 Sh./1996.
- Ibn Turka, *K. al-Mafāḥiṣ*, Tehran, Majlis Library, MS 10196, fols. 52a-118b.
- Ibn Turka, *Risāla-yi ḥurūf*, ed. Matthew Melvin-Koushki, in idem, “The Quest for a Universal Science: The Occult Philosophy of Šāʾin al-Dīn Turka Iṣfahānī (1369-1432) and Intellectual Millenarianism in Early Timurid Iran,” PhD dissertation, Yale University, 2012, pp. 462-74.
- Ibn Turka, *Sharḥ-i Naẓm al-durr*, ed. Akram Jūdī-Niʿmatī, Tehran: Mīrāth-i Maktūb, 1384sh/2005.
- Ikhwān al-Ṣafāʾ, *Rasāʾil*, 4 vols., Beirut: al-Aʿlamī, 2005.
- Ingenito, Domenico, *Beholding Beauty: Saʿdī of Shiraz and the Aesthetics of Desire in Medieval Persian Poetry*, Leiden: Brill, 2020.
- Killeen, Kevin and Peter J. Forshaw, eds., *The Word and the World: Biblical Exegesis and Early Modern Science*, New York: Palgrave Macmillan, 2007.
- Koch, Ebba, “The Taj Mahal: Architecture, Symbolism, and Urban Significance,” *Muqarnas* 22 (2005), pp. 128-49.
- Lange, Christian, “Le débat sur les mérites et les défauts de la vue et de l’ouïe dans l’histoire intellectuelle islamique,” *Bulletin des Études Orientales* 70 (2025), forthcoming.
- Melvin-Koushki, Matthew, “Mīr Dāmād’s *On Doffing*: Safavid Philosophy, Ritual Magic and Out-of-Body Experience,” forthcoming.
- Melvin-Koushki, Matthew, “Selenocentrism and Heliocentrism in Early Modern Persianate Imperial Cultures: ‘Alī versus Jesus, with Hermes Presiding,” *al-ʿUṣūr al-Wuṣṭā: Journal of Middle East Medievalists* 33 (2025), forthcoming.
- Melvin-Koushki, Matthew, “The New Brethren of Purity: Ibn Turka and the Renaissance of Pythagoreanism in the Early Modern Persian Cosmopolis,” *Intellectual History of the Islamicate World* 13.1 (2025), forthcoming.
- Melvin-Koushki, Matthew, “Timurid-Mughal Philosopher-Kings as Sultan-Scientists,” in *Rulers as Authors in the Islamic World: Knowledge, Authority and Legitimacy*, ed. Maribel Fierro, Sonja Brentjes and Tilman Seidensticker, Leiden: Brill, 2024, pp. 596-630.
- Melvin-Koushki, Matthew, “An Islamic Scientific Revolution? Early Modern Occult Science, Cosmic Philology and the Weird,” special roundtable issue, ed. Justin Stearns and Nahyan Fancy, *History of Science* 61.2 (2023), pp. 166-72.
- Melvin-Koushki, Matthew, “Better than Sufi Sex: Ibn Turka on the Superiority of Lettrism to Sufism as Model of Occult Islamic Humanism,” in *Islam ed esoterismo/Islam and Esotericism*, ed. Michele Olzi and Lisa Pizzighella, special issue of *La Rosa di Paracelso* no. 2 (2020 [2022]), pp. 53-80.
- Melvin-Koushki, Matthew, “Imperial Talismanic Love: Ibn Turka’s *Debate of Feast and Fight* (1426) as Philosophical Romance and Lettrist Mirror for Timurid Princes,” *Der Islam* 96.1 (2019), pp. 42-86.
- Melvin-Koushki, Matthew, “Of Islamic Grammatology: Ibn Turka’s Lettrist Metaphysics of Light,” *al-ʿUṣūr al-Wuṣṭā: Journal of Middle East Medievalists* 24 (2016), pp. 42-113.
- Montgomery, James E., *Al-Jāḥiẓ: In Praise of Books*, Edinburgh: Edinburgh University Press, 2013.

Piątak, Łukasz, “Between Philosophy, Mysticism and Magic: A Critical Edition of Occult Writings of and Attributed to Shihāb al-Dīn al-Suhrawardī (1156-1191),” PhD dissertation, University of Warsaw, 2018.

Selove, Emily, “Magic as Poetry, Poetry as Magic: A Fragment of Arabic Spells,” *Magic, Ritual and Witchcraft* 15.1 (2020), pp. 33-57.

Shaw, Gregory, *Hellenic Tantra: The Theurgic Platonism of Iamblichus*, Brooklyn: Angelico Press, 2024.

Uy II, Cyril V., *Lost in a Sea of Letters: Sa’d al-Dīn Ḥamūya and the Plurality of Sufi Knowledge*, Leiden: Brill, 2025.

## 2. Translation

*KM* = Ibn Turka, *K. al-Mafāḥiṣ*, Tehran, Majlis Library, MS 10196 fols. 52b-119a (copied 828/1425), with author’s autograph at beginning and end.

*RH* = Ibn Turka, *Risāla-yi ḥurūf*, ed. Matthew Melvin-Koushki, in idem, “The Quest for a Universal Science: The Occult Philosophy of Šā’in al-Dīn Turka Iṣfahānī (1369-1432) and Intellectual Millenarianism in Early Timurid Iran,” PhD dissertation, Yale University, 2012, pp. 462-74.

*ShND* = Ibn Turka, *Sharḥ-i Naẓm al-durr*, ed. Akram Jūdī-Ni‘matī, Tehran: Mīrāth-i Maktūb, 1384sh/2005.

### [§1. Opening of the Commentary] [= *ShND*, pp. 37-9]

*The palm of my eye served me heady lovewine  
my cup a face of one beyond beauty.*<sup>22</sup>

Love—reality’s foundation, saki to its feasthall—is the locus of all manifestation and manifesting (*zuhūr u iẓhār*) and the source of all perception and perceiving (*shu‘ūr u ish‘ār*). Receptivity’s (*qābiliyyat*) primacy leading inexorably to union (*jam‘iyyat*), it becomes the means whereby the cups of the Beloved’s coquetry are quaffed and the flagons of his/her perfect beauty drunk, to which love’s flush (*ḥumayyā ḥubb*) is an allusion. At every instant the human Eye’s (‘*ayn-i insānī*) faculty of sight—the Eye’s pupil (*insān-i ‘ayn*) being manifestation’s aperture (iris, *ḥadaqa*), nay, the spring (‘*ayn*) feeding the fruit of the tree of being’s garden (*ḥadīqa-yi vujūd*)—exerts itself in this office.

<sup>22</sup> The famous first line of Ibn al-Fāriḍ’s *Tā’iyya al-kubrā*: *saqatnī ḥumayyā l-ḥubbi rāḥatu muqlatī / wa-ka’sī muḥayyā man ‘ani l-ḥusni jallati*. The translation is by Homerin, 75. Under Ibn Turka’s pen, however, it is transformed from a classic Sufi anthem into scientific evidence for the supremacy of sight.



Eye tells heart the beloved's tale  
and has no view to keeping it from falling in love.

[Sa'dī]<sup>23</sup>

It is a Light compared to which the Sun, though lord of the realm of manifestation, is at this [cosmic] level but a servant in attendance at its court. The subtle realities have been dressed like brides in gross physicality and material substance for the purpose of realizing union: they are first decked out through its capable services with the jewelry of colors and tones, then presented in the Eyes' bazaar on the bridal throne of declaration till that the perceptive among them leads these realities—relative-singular of form (*hay'atī vaḥdānī-yi nisbī*) and essential-collective of image (*ṣūratī 'aynī-yi ijtimā'ī*)—to the courtyard of reception with the hand of its diffusive, all-embracing rays, there to present them for review by the chamberlains of the august palace of the Heart.<sup>24</sup>

I'll complain no more about my shredded heart  
since my pain's my eye's fault in all; [p. 38]  
This love wouldn't've wrecked my heart  
had my eye not seen such glory before it.

[Kamāl al-Dīn Ismā'īl]

In speaking of King Love (*salṭanat-panāh 'ishq*) [the poet] allows for the transcendence of the singular outward form (*ṣūrat-i vaḥdānī-yi vajhī*), free of all qualifiers and alluded to by the word “face” (*muḥayyā*), above that relative collective form (*hay'at-i majmū'ī-yi nisbī*) [assumed by subtle realities], expressed here by the word “beauty” (*ḥusn*).

[By contrast, these lines]:

Spake he Reality—and heard himself that speech  
showed himself—and saw himself his face

[‘Irāqī/Awḥad al-Dīn Kirmānī]

do not privilege the faculty of Sight but class it but one sense among others in the act of mediation, rather privileging the faculty of Hearing—but does this not imply a non-specificity of form [so perceived]? For however much the other senses may participate in the apperception of entities' forms (*ṣuvar-i ta'ayyunāt*) and the extrapolation of their underlying reality therefrom, only [Sight] can truly grasp that singular-collective manifest image of which all images are the expression. Most senses grasp only physical forms (*ṣuvar-i jismānī*), their material sub-

<sup>23</sup> Ibn Turka frequently quotes Sa'dī (d. 691/1292) as classical champion of cosmic-physical visuality; in the introduction (*ShND*, p. 6), for example, he opens his poetic valorization of the lettrist mode with another of his favorite Sa'dī lines: “No one reads the letter I read / no one knows the sign I know.” See Ingenito.

<sup>24</sup> This extramissionist simile of the heart as king and the physical senses as courtier informants echoes that of the Brethren of Purity in particular, despite Ibn Turka's disagreement as to their onto-epistemic hierarchy.

stances emptied and divorced from the subtleties of their spiritual bonds and occult realities. The Rational faculty itself, within whose compass intellected and posited objects certainly fall insofar as it is informed by all the senses, still remains incapable of comprehending them as they should be. For “[not every protected datum is divulged], nor can everything gazelles’ eyes dictate be transmitted”

Given Sight’s superior apperception of all such objects—in both their spiritual-divine (*rūḥāniyya*, *ilāhiyya*) and physical-existential (*jusmaniyya*, *kiyāniyya*) aspects, together with the singularity of their essential unicity (*aḥadiyyat jam‘iyyatihā l-dhātiyya*), the principle from which all forms arise—, whenever the conveyance and partaking of the winecup is spoken of it is this faculty that should be privileged.

Only a nightingale knows the measure of a rose yet to bloom—  
not all who scan bookleaves understand them.

[Ḥāfiz]

Although the faculty of Hearing has a wider compass of perception and operates on both physical and spiritual planes (*tarafayn*), precisely by reason of its bias toward the spiritual and its tendency toward aloof sublimity it is incapable of [p. 39] accurately rendering a form in its fullness, the mechanics of which were touched upon in the preface.<sup>25</sup> Since it is the nature of comprehensiveness (*jāmi‘iyyat*) to be the principle underlying all forms and the source of their essential receptivity, and given the fact of Hearing’s bias toward the spiritual and transcendent singlemindedness (*vaḥdat-i tanazzuh-nishān*), its status as a medium is in every case second to that of Sight; this order is thus observed in [Ibn al-Fāriḍ’s] ode. This also explains why he is obliged to use the feminine in every reference to the Beloved.

Since the intoxication of “love’s flush” (*ḥumayyā ḥubb*) (rising from the coverts of potentiality to the stage of actuality by means of Sight) severs external relationships and does away with phantasmic multiplicity (*kaṣrat-i vahmī*) and the distinguishing properties [of objects], this resulting in the emergence of true unicity (*vaḥdat-i ḥaqīqī*), in every case the point of reference of such acts of disassociation is absolute unicity (*vaḥdat-i muṭlaqa*), expressed here by the word “face” (*muḥayyā*). The relative distinction between objects—according to the simple relationship [expressed by the verse]

Saki, drinking partner, wine, tavern are one,  
candle, candlestick, flame, moth are one

and represented by the exact equivalence yet relative distinction which the letters of *ḥumayyā* (flush) have with *muḥayyā* (face)—references precisely this point.

<sup>25</sup> Cf. the ancient, more aurophile Pythagorean doctrine of the music of the celestial spheres as vehicle for Ascent.

[§2.] Introduction: On the Meaning of the Letter and Its Basic Divisions [= *RH*, pp. 464-5]

The root meaning of *ḥarf* (letter) in Arabic is the edge or end of something. In the present context it connotes a form sent down from the heaven of holiness that reveals all things—indeed, it also reveals things which have not entered into the circuit of existence, describing the manner of their [nonbeing]. For this reason the [concept of] *letter* must be considered more encompassing than [such concepts as] *existence* or *thing* or any other universal category that may be conceived. It therefore encompasses all things, comprehending each in precisely the same way without being more inclined to one than another; the letter thus remains standing straight [and unaffected] by virtue of its basic comprehensiveness. This is in contrast to everything else in existence, which inclines toward one of two opposing terms; thus *existence* is countered by *nonexistence*, and likewise with *thing*, [countered by *no-thing*], or *contingency*, [countered by *necessity*], or anything else that can be conceived. [Referring to the letter's primordial straightness and singularity, says He:] *Praise belongs to God Who has sent down upon His servant (‘abd) the Book, and has not assigned unto it any crookedness* (Q 18:1).

Now three loci of self-manifestation have been created for the letterform through which it manifests and reveals the end and the essence of every thing. The first is Sight (*baṣar*), to which the ‘ (‘*ayn*, Eye) in the word ‘*abd* (servant) refers; the second is Heart (*qalb*), to which the *B* in ‘*abd* refers; the third is Hearing (*sam‘*), to which the *D* in ‘*abd* refers. By this measure, then, the letter may be divided into three categories:

- 1) The *written-textual* form, which through the agency of fingers and hands is given form upon the open spread of white pages and reveals realities to both Sight and Insight (*baṣāyir*) as its proper loci; the exponents of this mode are *men of main and vision* (*ūlū l-aydī wa-l-abṣār*) (Q 38:45).
- 2) The *spoken-oral* form, which through the agency of the Tongue and the various points of articulation that modify the breath is embodied and expresses realities to Hearing and to Reason (‘*uqūl*) as its proper loci; its exponents are *those who make mention* (*ahl al-dhikr*) (Q 16:43, 21:7).<sup>26</sup>
- 3) The *spiritual-mental* form, which through the agency of the Rational (‘*āqila*) and Imaginal (*mutakhayyila*) faculties is analyzed within the broad realm of meaning with the Heart as its proper locus; its exponents are *men possessed of minds* (*ūlū l-albāb*) (Q 2:179, etc.); *He gives wisdom to whomever He will, and whoso is given wisdom has been given much good; yet none remembers save men possessed of minds* (Q 2:269).

---

<sup>26</sup> Which is to say, those who perform lettrist mantras—a primary theurgic practice from the Mongol era onward, when Indian yogic breath control and visualization (‘*ilm-i dam u vahm*) began to be systematically incorporated.

Each of these categories is specific to one of the three primary human faculties, to wit, Heart, Hearing and Sight. It is in this respect that quranic verses typically refer to all three together, usually giving precedence to either Heart (as in the verse *Surely in that there is a reminder to him who has a heart, or will give ear with a present mind* (Q 50:37), and the verse *There is nothing His like; He is the All-Hearing, the All-Seeing* (Q 42:11)) or to Hearing (as in the verse *And He appointed for you hearing and sight and hearts* (Q 16:78, etc.)). The first order reflects the fundamental and essential precedence of Heart with respect to the other members, and indeed with respect to all things in existence, whereas the second order reflects Hearing's precedence at the moment of creation, since it was the faculty singled out to receive the [spoken] command *Be!* (Q 2:117, etc.) from among the various members and faculties of perception.

However, because standard didactic usage gives precedence to that which is the most evident—as for example in the verse *How well He sees! How well He hears!* (Q 18:26)—, it is here more appropriate and useful to first treat of the written form of the letters. Indeed, the fact that the imperative form is used in the verse just cited suggests precisely the objective of instruction. Yet it must be noted that despite the fact that their written form is more manifest and their spiritual form more occult, the first is not self-evident and must be learned, whereas knowledge of the second need not be; that is to say, knowledge of the numbers and their degrees is innate, in contrast to knowledge of the written form of the letters and their shapes, which cannot be understood until they are learned. This is so because of a basic principle of Onemaking (*tawḥīd*), as those who have studied the subject know.

Each of them desired the quality  
 she had wrapped  
 in a form of loveliness shining forth  
 in a loveliness of form.<sup>27</sup> [p. 465]  
 [Ibn al-Fāriḍ]

Its beauty, occult, cannot manifest  
 yet is manifest through its occultness.

**[§ 3.] Codex Two: On the Textual Aspect of the Letter and Its Written Form Figuring Forth Occulted Meanings to the Coverts of Insight by Virtue of Its Manual Delivery onto the Folios of Sight [= KM, ff. 73a-74a]**

We have discussed at length the numerical, calculative principles that best establish philosophical verities with the clearest and most demonstrative of proofs. Let us now train our investigation on the written forms themselves, which figure forth those realities to direct visual percep-

---

<sup>27</sup> Ibn al-Fāriḍ, l. 244/246. For the translation see Homerin, 143.

tion [f. 73b] and clothe them in such manifest guises as to make them perceptible, rendering the most marvelous phenomena of all possible worlds perfectly expressible.

Know that the written form of the Letter is its most real and most perfect: whenever it would most fully manifest from out its normal occultation and ambiguity and undertake to ascend the towering peaks of exposition and lofty summits of disclosure, it [reveals itself to be] the most complete and eternal form of the Letter's very essence.

Now it may be argued that it is rather its spoken form that more justly holds this rank of perfection, given its famously exceptional role upon the pulpits of proclamation, the renown of its call in every ear and broadcast through the voices of every prayer tower; its ability to publish and make information perceptible far and wide applies not just to every [human] town and village but, uniquely, to all kinds of animals too. By contrast, the Letter's written forms can only be understood by an elite minority of humans, and that only after exhausting labors over an extensive period of time—and even then only through fresh application of vision and reflection in every new instance. Between its spoken and written forms, then, clearly there is no comparison with respect to making things perceptible and manifest.

I answer this argument by [classifying these two forms according to their respective] two modes. First, that of *summoning*: this is the tongue of prophethood (*nubuwwa*), whose conventional expressions are designed to be immediately comprehensible to a community in its entirety, exerting a powerful effect on the general populace and eliciting their full and devoted assent and commitment to the performance of bodily acts solely by means of verbal arguments and actions corresponding. It is obvious that speech alone holds the reins of this mode: its sharp rapier thrusts that quickly pass and are renewed with the renewing of each moment give it an exceptional ability to influence audiences' minds—hence the correct association of speech with this objective quite literally.<sup>28</sup>

Second, that of *guiding*: this is the tongue of sacral power (*walāya*),<sup>29</sup> whose specialized allusions have to do with matters more fundamental and can only be grasped by those of the necessary aptitude, kinship and patrimony, who win understanding thereof only to the degree they have achieved proximity [to the One] and the kinship that comes through inheritance: so do they continue to ascend the rungs of proximity on the ladder of meanings occult that lead to the Ultimate Object. It is likewise obvious that the written forms of the Letter alone hold the reins of this mode: through their bold lines and angles and planes and semicircles and circles that make up all letterforms both collectively and individually, noble conventions decked out like magnificent knights, they express [every thing and also every nothing] whatsoever. More: what they intimate perdures on the pages of days over years and eras, which perdurance alone makes it possible to understand—through diligent study and contemplation and personal veri-

---

<sup>28</sup> Hence the classic Platonic-scientific definition of poetry as "licit magic" (*sihr ḥalāl*). See Selove.

<sup>29</sup> Translated above as "sainthood," I find "sacral power" for *walāya* more appropriate to its early modern Persianate imperial usage generally and Timurid specifically, whereby Ibn Turka's promotion of *saint-philosopher-kinship* and valorization of the Moderns over the Ancients while marrying the two became all the political rage.

fication—an operative portion of revealed verities, which splits open the chest and so makes easily knowable those things not contained in any books.

Herein is a subtle but pivotal point that speaks to the peculiar nobility of the written form of the Letter: it alone can intermix all its percepts, unlike every other sensible form. For its medium is Light, which conveys it to the Light of Sight—and Light combined with Light does not simply juxtapose percepts but intermixes them perfectly. Thus clashing colors can be combined in a single apperception, and no color is negated by its opposite.<sup>30</sup> This is in contrast to the percepts of all other sensory faculties, including Hearing, Smell, Touch and Taste, which denature each other when combined, since their medium is air or a similar body; naturally, such mediums can only properly transmit clashing forms [to the senses] one at a time. But not Light, that medium of the written form: since it is not a body, clashing forms cannot jostle within it [f. 74a], which makes Sight unique among the senses in its ability to transcend the denaturings and dark defilements of materiality. Hearing—trained upon transitory accidents borne by sound-waves that pass with each instant’s passing—can hardly compare.

The more descended forms are, the more they are perfect, universal and encompassing of special properties. Written forms are manifestly lower than spoken ones, to which they correspond, just as spoken forms correspond to [higher] mental ones; they are therefore [the only forms] that totally and utterly descend—while simultaneously totally and utterly ascending. In their comprehensiveness and peculiar ability to unify all opposites within perception they function like the Heart, which perceives the same. They are thus Heart’s form [and vehicle] in the realm of perception. Any who would understand this [correspondence] directly need only numerically analyze the *bayyināt* of [the word] *Book* (*kitāb* > *KTAB* = 423 >> *YMAFA* = 132), which perfectly expresses the principle.<sup>31</sup>

Surely you see that whenever the Sealing Revelation (*tanzīl*, lit. “Downsending”) would glorify the letter generally and expound its universal principles without specification it always refers to itself as *The Book*? One need only consider that first sent down to the Seal as a message to all, to wit, the saying of the Most High: *Read: In the Name of your Lord Who created, created Man of a bloodclot. Read: And your Lord is the Most Generous, who taught by the Pen, taught Man what he knew not* (Q 96:1-5). Thus the glorious Revelation itself 1) opens with the command to read; 2) informs the human being of his created and therefore primordially subject status; 3) sigills it by repeating the opening command and swearing by the *Most Generous Lord*; 4) defines

<sup>30</sup> This point echoes earlier theological discussions; see e.g. *ISH*, vol. 2, ch. 26 (al-Nazzām), §1.6; ch. 27 (al-Ash‘arī on al-Jāhiz and others), §3.6ff.

<sup>31</sup> When performing *taksīr* (cognate to Hebrew *temurah*), which involves the separation of the letters of a name or word and the writing out of the letternames in full, then the elimination of repeated letters (e.g., Aḥmad > AḤMD > ALFḤAMYMDAL > ALFḤMYDL), the term *zubur* refers to the first letters in the full letternames (e.g., the A in ALF) and *bayyināt* to the remaining letters (e.g., LF in ALF); both terms derive from Q 3:184, 16:44, 35:25. This abbreviated calculation in the margin begins with the root value of *Book* (*kitāb* > *KTAB*) as 423, whose *bayyināt* after several cycles of reduction (not specified in the manuscripts available to me) reveal the essential Number of *Book* to be 132; that this is the Number of *Heart* (*qalb* > *QLB*) too is conclusive proof of their existential, primordial identity.

that *Lord* by His use of the Pen to make known [all things]; and 5) elevates the essentially subject human by means of this Pen teaching—teaching what he could not know but by that means.<sup>32</sup> No affirmation of the superiority of the written aspect of the Letter could be more patent or manifest.

By the same token, whenever He would refer to a counterpart of either Wisdom (*ḥikma*) or Faith (*īmān*) that encompasses all its own opposites he refers to *The Book*. For there are two modes for attaining scientific knowledge of reality, regardless of the extreme specialization of all its many disciplines and branches. In the first mode, such certain knowledge is attained through the very essence of the Heart itself, whether through definitive and demonstrative observation (*anẓār*) or through brilliant and ecstatic flashes of intuition (*adhwāq*); in its specific sense this mode is termed Philosophy and in its general sense Faith.<sup>33</sup> In the second mode, it is attained through the bare manifestations of the letters themselves [visually perceived], which [numerically] reveal the essence of all things from their origin to end—which revelation is called *The Book*, referring to the [cosmic] whole by its most iconic and perfect and perfectly manifesting part.

---

<sup>32</sup> Given the author's mathematizing program, that the verbal noun *ta'lim* is repeated here is likely meant to resonate with *ta'ālīm*, the standard Arabic term for the Pythagorean quadrivium.

<sup>33</sup> This is Ibn Turka's elegant solution to and transcendence of the hoary Science vs. Religion binary: both then and now, it is well established that empirical observation and dreamtime intuition are equally important to the business of Science, whose sociopolitical manifestations are often indistinguishable from Religion—especially when the natives are marked for violent colonization. But like Lord Love, the Letter marries all epistemic opposites to make possible and authorize every form of human knowing and doing and seeing.